



ADDING THE GREEN LAYER TO THE HEAT TWIN

The green layer.

Which vegetation datasets we're using for the twin, and which of the sources in your email have **ended** or turned out to be **non-commercial**.

"Your three-tier plan was right. Three of the named sources won't survive contact with our licence. Here's what does."

DATE	PREPARED FOR	FOCUS	SOURCES
10 August 2026	The co-founders	Vegetation sourcing	Primary-verified

01 • EXECUTIVE SUMMARY

The plan is right. Some of the **sources** aren't.

Vegetation is the biggest physical term the twin is still missing. Shade and evapotranspiration are how a hot city actually cools itself. Your three-tier email (a raster for correlation, canopy height for shade, individual trees for realism) is the correct way to model it, and we're keeping that structure exactly.

The catch is licensing and end-of-life. We're a commercial company, so "free to download" isn't the test. "Free to use in a product we sell and pitch" is. When we verified each named source against its primary licence and status, three of them failed that test, and one needs imagery we can't legally get over an Indian city.

What we're using (all commercial-clean): **Sentinel-2** for the NDVI raster (already ours), the **Meta / WRI 1 m Canopy Height model** for the shade term (the one genuinely new ingest, CC BY 4.0, anonymous AWS, no Google Earth Engine), **ESA WorldCover** as a land-cover mask (already ours), and **procedurally-generated trees** for the 3D layer.

What we can't use, with the reason each was cut: **NICFI** (programme ended Jan 2025, and was non-commercial anyway), **Google Earth Engine free tier** and the **Dynamic World / EIE** workflows that ride on it (non-commercial; commercial needs paid EE), **FABDEM** (the free tier is non-commercial), and **DeepForest** (a fine tool, but it needs sub-metre imagery we don't have).

THE ASK

Sign off on ingesting the **Meta/WRI 1 m Canopy Height** model as Tier 2 and generating trees **procedurally** for Tier 3. Everything else in the green layer is already commercial-clean and shipping. No source in this plan depends on a dead or non-commercial dataset.

02 · THE VEGETATION STACK WE'RE USING

Three tiers, all commercial-clean.

Your tiering, kept intact. The only changes are which dataset fills each tier, chosen so every one is licensed for a product we sell. Each is primary-verified (see Sources).

Tier 1 · NDVI raster · Sentinel-2 L2A · already ours

COMMERCIAL ✓

Copernicus Sentinel-2 surface reflectance via the earth-search STAC API, the same pipeline already in the twin. NDVI at 10 m, ~5-day revisit, used as the thermal correlator. Copernicus is **free, full & open, with commercial use explicitly permitted**. That's the best licence available to us.

STATUS → Shipped. No change. This tier is done.

Tier 2 · Canopy height · Meta / WRI 1 m Canopy Height (v2) · the one new ingest

COMMERCIAL ✓

Per-pixel canopy height at **1 m**, the actual shade term rather than just greenness. On **AWS Open Data**, anonymous, no credentials, **no Google Earth Engine**. Licence: **CC BY 4.0** (commercial with attribution). Honest limits we'll receipt: single-epoch (~2018–2020), MAE of a few metres, cloud gaps, occasional ~150 m tiling artefacts. Good as a shade/cooling input, not a survey.

ACTION → Ingest as the Tier-2 canopy layer. This is the sign-off ask.

Tier 2 · Land-cover mask · ESA WorldCover 10 m · already ours

COMMERCIAL ✓

Tree / grass / built classes at 10 m (CC BY 4.0). Cross-checks and gap-fills the canopy model where it has cloud holes.

STATUS → Already in the pipeline. Reused, not re-fetched.

Tier 3 · Individual trees · procedurally generated · CHM-driven

MODELLED · RECEIPTED

Reference-shaped broadleaf + palm, instanced in three.js, **placed and scaled from the Tier-2 canopy heights**. Rendered as clearly-illustrative geometry and labelled **modelled, not measured**, in keeping with the twin's receipts posture. Why not buy 3D tree models: the marketplaces (RenderHub, Sketchfab) return almost only North-American / East-Asian species, with no clean commercially-licensed South-Asian street trees (neem, gulmohar, krishnachura, palms). A wrong-species model would be **less** honest than an obviously-schematic one.

ACTION → Generate procedurally. Fallback if ever needed: Kenney/Quaternius **CC0** models (clean but generic).

03 · THE SOURCES THAT DIDN'T SURVIVE VERIFICATION

Great leads, stopped by **licence** or **end-of-life**.

These were all reasonable to try; the option-space in your email was the right one. Each rejection below is an **existence or licence fact** checked against the primary source, not a quality judgement.

Planet NICFI basemaps

ENDED + NON-COMMERCIAL

The NICFI Satellite Data Program **wound down in Jan 2025**; the replacement tender was **cancelled in Sept 2025**, so there is no live successor. And even when live, NICFI access was **non-commercial / restricted**, never valid for a for-profit product.^[4]

CORRECTION ON THE RECORD → I'd earlier said NICFI "didn't cover India." That was wrong: its 30°N-30°S band **did** include Kolkata. The verdict is unchanged (dead + non-commercial), but for the right reason.

Google Earth Engine (free tier), and Dynamic World / EIE on top of it

NON-COMMERCIAL

The email's NDVI, Dynamic World and Environmental-Insights workflows all route through GEE. **Free GEE is non-commercial only**; a for-profit needs a **paid Earth Engine** licence.^[5] The underlying data (Dynamic World, EIE canopy) is open, but there's **no clean bulk access outside GEE**, so using it means the paid licence.^[6]

HOW WE DODGE IT → We pull EO through open STAC + AWS Open Data (earth-search, Planetary Computer, dataforgood): same data, commercial-clean, no EE bill.

FABDEM (free tier)

NON-COMMERCIAL

Suggested as the surface to drape canopy onto. The **free FABDEM is CC BY-NC-SA 4.0, which is non-commercial**. A paid commercial licence exists (via Fathom) but isn't free.^[7] For terrain we already use commercial-clean sources (terrarium / Copernicus GLO-30).

NOT NEEDED → Our terrain path is already licensed correctly; no FABDEM required.

DeepForest (individual-tree detection)

BLOCKED UPSTREAM

DeepForest is **MIT-licensed and perfectly good**. The blocker is the input it needs: **sub-metre RGB (~10 cm)** to find tree crowns. We have **no legal sub-metre imagery over Kolkata** (no drones over Indian cities; no licensed aerial tiles). The email's "as you did with SAM for buildings" doesn't apply, because our footprints came from **Overture**, not our own high-res imagery.^[8]

WHY PROCEDURAL WINS → With no imagery to run detection on, CHM-driven procedural trees are the honest Tier-3.

04 · THE NEAR-MISSES, AND ONE LINE WE WON'T CROSS

Usable licence, wrong fit, and the claim we refuse.

A few more sources are commercially clean but don't earn a tier on the merits.

Recording them so we don't re-litigate later.

SOURCE	LICENCE	WHY IT'S NOT A TIER
WRI GFW Tropical Tree Cover	CC BY 4.0 ✓	Kolkata sits at the northern edge of its tropical extent, and it's coarser than the Meta CHM. Cross-check only.
OSM NATURAL=TREE	ODbL ✓	Only ~464 tree nodes across our wards, far too sparse to drive canopy density.
Mapillary	CC-BY-SA	Street-level imagery, but the API exposes no tree/vegetation class , plus share-alike + anti-scrape terms. ^[9]
KMC tree census	n/a	No open, machine-readable municipal tree inventory for Kolkata exists.

The one claim we won't make

No AQI / pollution-dispersion layer, no wind-CFD. The twin's physics is **2-D: SUN** and **KRAD** are scalars, and there's no advection or 3-D wind field. Bolting an air-quality or dispersion story onto a model that can't represent it is exactly the "impressive but hollow" failure we're differentiating against.

Vegetation enters the model as **shade and a cooling offset**, terms the 2-D physics can honestly carry. Anything that needs a wind field, we don't claim.

THE HONESTY LINE FOR THE GREEN LAYER

This keeps the green layer on the right side of our whole thesis: we surface what we can defend, and we say plainly what we can't.

05 • NET RESULT & SOURCES

One new ingest. Zero licence traps.

TIER	SUGGESTED IN THE EMAIL	WHAT WE SHIP	LICENCE
1 · NDVI	GEE Sentinel-2 / NICFI	Sentinel-2 L2A (earth-search STAC)	Copernicus ✓
2 · canopy	Google EIE / Dynamic World / FABDEM drape	Meta/WRI 1 m CHM + ESA WorldCover	CC BY 4.0 ✓
3 · trees	DeepForest + CesiumJS	Procedural, CHM-driven, receipted	Own geometry ✓

Two of three tiers are already shipping and commercial-clean. The single new ingest is the **Meta/WRI 1 m canopy-height model**. Trees are ours. Nothing depends on a dead or non-commercial source.

SOURCES • PRIMARY-VERIFIED, 10 AUGUST 2026

1	Sentinel-2 / Copernicus open licence (commercial permitted) · sentinels.copernicus.eu · earth-search STAC · earth-search.ch.aws.element84.com	✓ USING
2	Meta / WRI 1 m Canopy Height (v2), AWS Open Data (CC BY 4.0) · registry.opendata.aws/dataforgood-fb-forests · Meta announcement	✓ USING
3	ESA WorldCover (CC BY 4.0) · esa-worldcover.org · AWS · registry.opendata.aws/esa-worldcover	✓ USING
4	Planet NICFI, wind-down Jan 2025; programme page · planet.com/nicfi · nicfi.no	✗ ENDED · NC
5	Google Earth Engine terms, free tier non-commercial; commercial needs paid EE · earthengine.google.com/commercial	✗ NON-COMM
6	Google Environmental Insights Explorer · insights.sustainability.google · Dynamic World · dynamicworld.app (no clean non-GEE bulk access)	✗ GEE-BOUND
7	FABDEM, free tier CC BY-NC-SA 4.0 · data.bris.ac.uk · commercial via Fathom · fathom.global	✗ NON-COMM
8	DeepForest (MIT; needs sub-metre RGB) · deepforest.readthedocs.io	△ NO IMAGERY
9	Mapillary API / terms (no tree class; CC-BY-SA) · mapillary.com/developer	✗ NO CLASS
10	WRI GFW Tropical Tree Cover (CC BY 4.0) · data.globalforestwatch.org · OSM ODbL · openstreetmap.org/copyright · Kenney CC0 · kenney.nl/assets	🔍 CROSS-CHECK